Unit 6.2 Vectors (p.798)

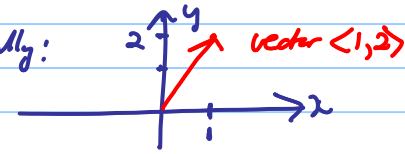
Note Title 5/21/2018

A vector

- has magnitude and direction
- does not have a fixed position.

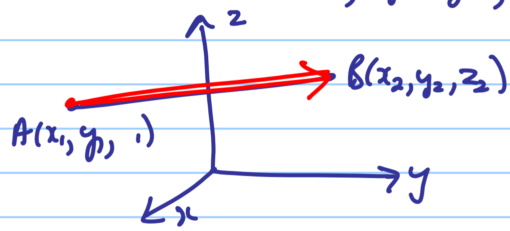
① Present vectors geometrically:

$\langle 1,2\rangle$ in $\mathbb{R}^2$:



② Vector between 2 points $A(x_1,y_1,z_1)$ and $B(x_2,y_2,z_2)$

is: $\overline{AB} = \langle x_2-x_1,\ y_2-y_1,\ z_2-z_1\rangle$

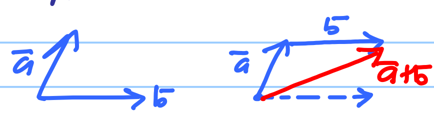


$\overline{AB}$: directed line segment with endpoint B and starting point A.

See Example 3 on page 800.

- Operations with vectors: (p. 799)

① Addition. Suppose $\bar{a}$ and $\bar{b}$ are in $\mathbb{R}^3$

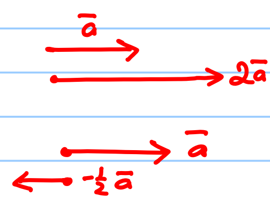


Triangle law.

② Scalar multiplication: $c\bar{a}$, $c\in\mathbb{R}$

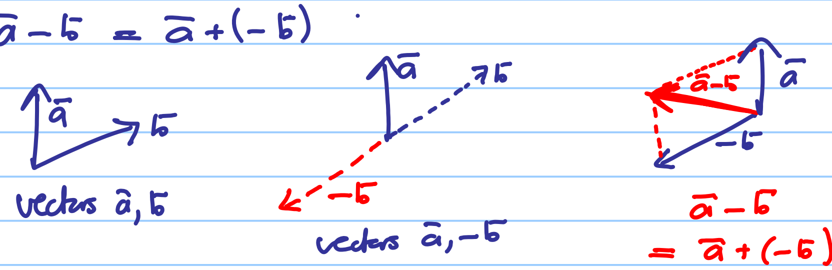
$c>0$, e.g. $c=2$

$c<0$, e.g. $c=-\frac{1}{2}$



③ Subtraction:

$\bar{a}-\bar{b} = \bar{a}+(-\bar{b})$



- Properties of vectors: NB (See p. 802.)

- $\bar{a}$ and $\bar{b}$ are parallel if there exists a $c\in\mathbb{R}$, such that $\bar{a}=c\bar{b}$. Same direction: $c>0$.

- The length / norm of a vector $\bar{a}$ is defined as:

$$|\bar{a}| = \sqrt{a_1^2 + a_2^2} \quad \text{if} \quad \bar{a} = \langle a_1, a_2 \rangle$$
$$|\bar{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2} \quad \text{if} \quad \bar{a} = \langle a_1, a_2, a_3 \rangle.$$

- A unit vector is a vector with length 1.

p. 803 If $\bar{a} \neq \bar{0}$, then $\frac{\bar{a}}{|\bar{a}|}$ is a unit vector in the direction of $\bar{a}$.

- The standard basis vectors are:

p. 802 in $\mathbb{R}^2$: $\bar{i} = \langle 1, 0 \rangle$ and $\bar{j} = \langle 0, 1 \rangle$

in $\mathbb{R}^3$: $\bar{i} = \langle 1, 0, 0 \rangle$, $\bar{j} = \langle 0, 1, 0 \rangle$ and $\bar{k} = \langle 0, 0, 1 \rangle$.

Any vector can be written in terms of the standard basis vectors, e.g. $\bar{a} = \langle a_1, a_2, a_3 \rangle$ in $\mathbb{R}^3$ can be written as:

$$\begin{aligned} \bar{a} &= \langle a_1, a_2, a_3 \rangle \\ &= \langle a_1, 0, 0 \rangle + \langle 0, a_2, 0 \rangle + \langle 0, 0, a_3 \rangle \\ &= a_1 \langle 1, 0, 0 \rangle + a_2 \langle 0, 1, 0 \rangle + a_3 \langle 0, 0, 1 \rangle \\ &= a_1 \bar{i} + a_2 \bar{j} + a_3 \bar{k}. \end{aligned}$$

- Note that it is important to use the notation of the textbook.
In the textbook a vector is printed in BOLD, but, since we cannot write in bold, we make a bar to indicate we work with a vector:

$$\bar{a} = \langle a_1, a_2, a_3 \rangle$$

Some examples:

(1) Find $2\overline{AB} - 3\overline{AC}$ if $A(2,-1,0)$ and $B(-2,4,1)$ are given.

(endpoint) − (starting pt)

$$\overline{AB} = \langle -2, 4, 1 \rangle - \langle 2, -1, 0 \rangle = \langle -4, 5, 1 \rangle$$
$$\overline{AC} = \langle 1, 2, -2 \rangle - \langle 2, -1, 0 \rangle = \langle -1, 3, -2 \rangle$$

$$2\overline{AB} - 3\overline{AC} = 2\langle -4, 5, 1 \rangle - 3\langle -1, 3, -2 \rangle = \langle -5, 1, 7 \rangle$$

(2) $A(0,1)$, $B(1,1)$, $C(0,2)$ and $D(3,2)$ are points in $\mathbb{R}^2$.
(a) Determine if $\overline{AB}$ and $\overline{CD}$ are parallel.
(b) " " $\overline{AC}$ and $\overline{BD}$ " " .

(a) $\overline{AB} = \langle 1, 1 \rangle - \langle 0, 1 \rangle = \langle 1, 0 \rangle$
$\overline{CD} = \langle 3, 2 \rangle - \langle 0, 2 \rangle = \langle 3, 0 \rangle$

Since $\langle 3, 0 \rangle = 3\langle 1, 0 \rangle \Rightarrow \overline{AB}$ and $\overline{CD}$ are parallel.

(b) $\overline{AC} = \langle 0, 1 \rangle$
$\overline{BD} = \langle 3, 2 \rangle - \langle 1, 1 \rangle = \langle 2, 1 \rangle$
$\Rightarrow$ 2 vectors are not parallel,

since there is no $c \in \mathbb{R}$ such that $\overline{AC} = c\,\overline{BD}$.

(3) $\langle 1,2,-2\rangle$ is parallel to $\langle 4,8,-8\rangle$

$(\langle 1,2,-2\rangle \parallel \langle 4,8,-8\rangle)$, since $\langle 1,2,-2\rangle = \frac{1}{4}\langle 4,8,-8\rangle$.

(4) $\langle 1,2,-2\rangle \parallel \langle -\frac{1}{2},-1,1\rangle$, since

$\langle 1,2,-2\rangle = -2\langle -\frac{1}{2},-1,1\rangle$.

(5) A unit vector in the direction of $\bar{a} = \langle -3,4\rangle$, is:

$$\frac{\bar{a}}{|\bar{a}|} = \frac{\langle -3,4\rangle}{|\langle -3,4\rangle|} = \frac{\langle -3,4\rangle}{\sqrt{9+16}} = \frac{1}{5}\langle -3,4\rangle = \langle \frac{-3}{5},\frac{4}{5}\rangle.$$

(6) Find a vector $\bar{b}$ with length 2 in the direction opposite to $\langle -3,4\rangle$.

$\bar{b} = -2\langle \frac{-3}{5},\frac{4}{5}\rangle$ (from (5))

($-$ for opposite direction)